

Single-Cell Li-Ion 3.3V / 500mA Low-Noise Power Supply

Professional GitHub README Documentation

Project Overview

A compact, low-noise, and highly protected 3.3V power supply designed for battery-powered embedded applications. The design converts a single-cell Li-Ion battery output into a regulated 3.3V rail capable of supplying up to 500mA continuous current.

Key Specifications

Input Voltage: 3.8V–4.2V DC

Output Voltage: 3.3V DC

Maximum Output Current: 500mA

Typical Load Current: 450mA

PCB Layers: 2

Operating Temperature: 0°C–70°C

Board Size: < 40mm × 20mm

Applications

Embedded Systems, IoT Devices, Wireless Sensor Nodes, RF Modules, Portable Electronics, Low-Noise Analog Circuits, Battery-Powered Development Platforms.

Hardware Architecture

Battery Input → PTC Fuse → TVS Protection → Reverse Polarity MOSFET → LP5912 LDO Regulator → Output Filtering → 3.3V Output

Major Components

LP5912-3.3DRVR Low-Noise LDO Regulator, DMP2035U-7 P-Channel MOSFET, MF-NSMF110-2 Resettable Fuse, SMAJ5.0CA TVS Diode, BZT52C6V8 Zener Diode.

Protection Features

Reverse polarity protection, overcurrent protection, surge suppression, ESD protection, and MOSFET gate protection.

Filtering and Stability

22 μ F and 100nF input capacitors provide bulk storage and decoupling. 10 μ F and 100nF output capacitors ensure stability and transient performance.

Thermal Design

Worst-case regulator dissipation is approximately 0.45W. Thermal management is achieved using an exposed pad, thermal vias, ground stitching vias, and a continuous ground plane.

Test Points

TP1: VIN, TP2: VOUT, TP3: GND.

PCB Layout Features

2-layer PCB, continuous ground plane, thermal via array, wide power traces, short current paths, 45-degree routing, and clear silkscreen labeling.

Project Status

Schematic Complete, PCB Layout Complete, ERC Passed, DRC Passed, Manufacturing Files Generated, Ready for Fabrication.

Repository Contents

schematic/, Layout/, Gerber files/, Drill files/, BOM/, Datasheets/, Design Notes/, 3D_Models/, symbols/, footprints/, README.md

Tools Used

KiCad 10, Manufacturer Datasheets, Standard 2-Layer PCB Design Practices.

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